INTENT · ENGINEER AND CLARIFICATION

a Brief, actors and tools

traced example: the recorded b06 agent session

b06 Engineering brief

"Plate envelope 160 x 60 x 120 mm (x, y, z), y up, aluminium 6061 … four M10 bolts … a 24 mm washer. Keep a solid boss … a 40 x 40 mm, 5 mm thick solid pad … carries 2 kN downward and 0.8 kN sideways (+x) … Volume fraction 0.2, 1 mm elements."

suite: 50 briefs; 40 to solve, 7 needing a question, 3 to refuse

GUI Engineer in the GUI

three.js viewport: pick faces, patches and edges; place keep-in and keep-out regions; accept or reject agent cards

LLM Language agent over MCP

two model providers:
Codex gpt-6.1-sol (primary), Claude
Code claude-opus-5-5 · Propose or Act

One logged tool contract 25 tools

edit 12	new_spec, set_params, …
ground 4	validate_spec, render, …
ask 1	ask_clarification
compute 6	ladder, solve, status, field_stats, …
export 2	export_stl, export_report

GUI and agent call the same tools; the log keeps actor, arguments, spec hash **and outcome; no tool writes a certificate**

b Shared specification

one typed object, edited by GUI and agent through the same tools

LLM Clarification

2–4 concrete readings when a missing fact changes the design. In b06:
"Does '20% material' mean 20% of the optimizable volume … or 20% of the complete envelope including those solids?"

v* = 0.2 of design elements ✓

20% of envelope: v* = 0.181

oracle answer logged, applied by set_params

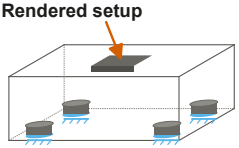
50 briefs right: 44 with tools → 50 with questions (p = 0.031); b06: 4 questions

Typed specification agentto.spec/v0

units	mm, N
domain	box 160 × 60 × 120
material	E = 69 000 MPa, Poisson ratio 0.33
volume	v* = 0.2 of the design elements
supports	4 washer patches on y−, fix x, y, z
load	(800, −2000, 0) N on a y+ patch
keep-in	4 bosses Ø24 × 10 mm, 1 pad (solid)
resolution	h = 1 mm (base request)

face · patch · edge · box · cylinder

Rendered setup



validate at h = 1 mm: 1,126,080 design elements, no rule findings

Picture check

3 views → reviewer → repair; advisory, never a gate

validate ↓ · error: edit, revalidate ↑

RULES

Gate 1 Rule base

13 deterministic checks; errors block solving and release
filled: error · open: warning

spec R001 R002 R004 R006 R007
physics R003 R005 R008 solver R012 R013
geometry R010 R011 resolution R009

With the gate removed

a load inside a keep-out region converged and was certified at 3.8×10^{8} × the reference compliance (R008); 6 wrong G specs passed

every rule: intent is not a rule property

NUMERICS · SOLVER CERTIFICATE

c Resolution ladder

warm, coarse to fine, each rung started on request; renders of this session's fields

sketch h = 2 mm



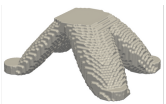
80 × 30 × 60 = 0.14M
cold, 80 iterations, 6.4 s
feedback only, uncertified

preview h = 1 mm



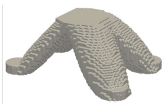
160 × 60 × 120 = 1.2M
warm, 12 iterations, 1.9 s
feedback only, uncertified

refine h = 0.5 mm



320 × 120 × 240 = 9.2M
warm, 12 iterations, 9.1 s
feedback only, uncertified

release h = 0.25 mm



640 × 240 × 480 = 73.7M
warm, 12 iterations, 35.4 s
spill guard on → Gate 2

Release schedules vs cold fine runs: 5 problems, 25.6–73.7M elements

schedule	settings after transfer	compliance vs cold	release-rung compute
tail	p 4.5, β 32, move 0.05	+0.005% to +1.39%	3–11% of cold
long re-open	from p 3.5, β 4, move 0.15	−3.06% to +0.07%	51–69% of cold

tail certifies the transferred design; long re-opening re-optimizes it on the fine grid
warm transfer: upsample ρ, one filter pass, bisect a density offset to meet v*

Solver

matrix-free voxel FEM, multigrid PCG; mixed-precision refinement at release; RTX 4090 with 24 GB

d Solver-issued certificate

Gate 2, issued by the solver for the release rung

What the gate caught

486 recorded solves; of 54 release attempts 52 were certified, 2 gave no releasable result; 11 unconverged: none released
spill guard stopped two 113.2M runs and one 59M evaluation

Six predicates, all required

$A_{num} = A_{conv} \wedge A_{res} \wedge A_{no\,spill} \wedge A_{monitor} \wedge A_{vol} \wedge A_{passive}$

- ✓ **converged** — optimization met its stop test
- ✓ **state solve accepted** — $r_{true} = \|f - Ku\|_2 / \|f\|_2 \le \tau = 10^{-5}$ in FP64
- ✓ **no VRAM spill** — none observed during the release solve
- ✓ **spill monitored** — GPU memory counters sampled in-process
- ✓ **volume** — max error ≤ 10^{-3}; b06: 0.2000
- ✓ **passive exact** — ρ = 1 in keep-in, 0 in keep-out regions

Release needs Gate 1 and Gate 2; uncertified fields stay feedback, and agent text cannot substitute for either

RECORD

e Release and record

Certified design



73,728,000 elements
ρ = 0.5 isosurface, VTK Flying Edges

Exports

STL: 1,595,756 triangles, closed; 0 boundary, 0 non-manifold edges
report: six checks, grid, units, compliance 4.011 N·mm, spec hash c9bd97c53d93; an edit after release marks the result stale, never current

Session log and replay

36 tool calls, 0 failed; 4 questions; 176 s logged, 201 s CLI wall time
replay without the language model: 36 calls, 5 solves, 0 mismatches
15 sessions: 14 released, 1 refused (R003); all 15 replayed exactly